ETHAN LARKIN

Software Engineer

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ABOUT ME

Associate Software Engineer at L3Harris on the Fusion ELINT team, working in C++, Python and Bash across message systems (NATS, protobuf, FlatBuffers), containerized microservices, GitLab CI/CD and test automation. UT Dallas computer science graduate who worked the whole way through school, and who likes turning command-line tools and manual processes into something anyone can use.

EXPERIENCE

L3Harris Greenville, TX

Associate Software Engineer, Fusion ELINT 2025 – Present

- Wrote a C++ daemon that migrated 15 message types from a legacy internal messaging system to NATS and protobuf
- Built GitLab CI/CD pipelines that package the system's ICD (message definitions) as RPMs, replacing a hand-maintained copy in each microservice with a single source installed wherever it's needed
- Expanded test infrastructure with 50+ pytest-bdd scenarios that catch errors before they reach production
- Built 3 web GUIs (HTML, CSS, JavaScript) over existing command-line tools so non-technical users can run them without a terminal, with fewer input errors and no engineer support
- Deploy the subsystem's microservices together with Podman, and built a Kubernetes software set that deploys several analyst tools together
- Designed a standardized FlatBuffers message readable by anyone with the message definition; use SQLite in a self-contained microservice and Bash for subsystem scripting
- Use AI tools throughout the development workflow, cutting estimated bug-fix time by about 40%

Earlier work (2018 – 2025, through school): QuikTrip, Emerald Cottages, Whataburger

PROJECTS

Drop a Bop — *TypeScript, React, AWS Lambda, DynamoDB, AWS SAM* Fall 2026

- Private music-sharing app where friend groups share a song each weekday, rate the week's songs, and see results, stats and leaderboards
- Hosted on Cloudflare Pages with a serverless AWS back end (Lambda, API Gateway, DynamoDB, Cognito) defined in AWS SAM; runs at extremely low cost with ~1,000 automated tests
- Built with AI assistance (Claude Code): I set the direction and reviewed each step. Try the demo at dropabop-demo.pages.dev

Direction-Finding Drone — *L3Harris-sponsored senior design · C++, CUDA, MATLAB, Jetson Nano* Fall 2024

- Helped design the original FFT processing in MATLAB, then wrote parallel CUDA FFT code in C++ on a Jetson Nano to process SDR-captured signals in near real time
- Led the move to CUDA unified memory so the CPU and GPU share data instead of copying it, cutting compute time
- Helped connect FFT output to a live, real-time spectrogram; 5-person CS team working with a 6-person CE team

Banking Application — *Next.js, TypeScript, Python, Django, PostgreSQL, Azure* Fall 2024

- Set up automatic deployment to Azure Static Web Apps with GitHub Actions and connected the Django backend to Azure-hosted PostgreSQL
- Designed the database models for members, bank accounts and transactions in a 5-person team app with RSA-encrypted transaction details and hashed credentials

EDUCATION

University of Texas at Dallas — B.S. Computer Science *(Richardson, TX)* December 2025

SKILLS

Languages	C++, C, Python, Bash, CUDA, JavaScript, TypeScript, Java, SQL, MATLAB, HTML/CSS
Messaging & Data	NATS, Protocol Buffers, FlatBuffers, PostgreSQL, SQLite
DevOps & Cloud	GitLab CI/CD, GitHub Actions, Podman, Docker, Kubernetes, RPM, Linux, AWS, Azure
Frameworks & Testing	React, Next.js, Django, pytest-bdd